Energy in Simple Harmonic Motion

$$E = \tfrac{1}{2}mv_x^2 + \tfrac{1}{2}kx^2 = \tfrac{1}{2}m\omega^2A^2\sin^2(\omega t + \phi) + \tfrac{1}{2}kA^2\cos^2(\omega t + \phi) = \tfrac{1}{2}kA^2$$

$$\boxed{k = \frac{2\pi}{\lambda}} = \frac{2\pi f}{\lambda f} = \frac{\omega}{v}$$

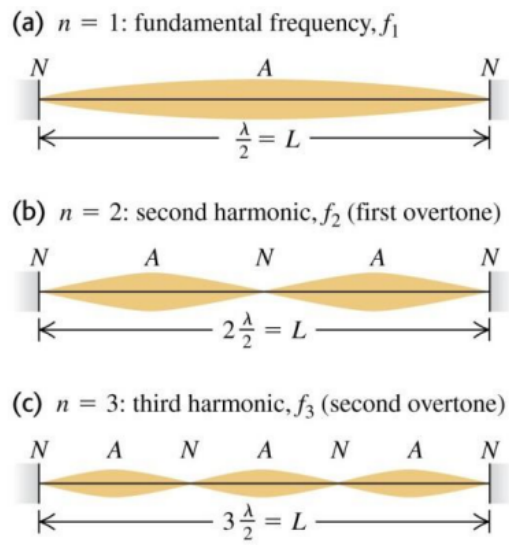
$$P_{\text{av}} = \frac{1}{2}\sqrt{\mu F}\,\omega^2A^2 = \frac{1}{2}P_{\text{max}}$$

Normal mode frequencies are

$$L = n\frac{\lambda}{2} \quad \Rightarrow \quad \lambda_n = \frac{2L}{n},$$
$$n = 1, 2, \ldots$$

and frequencies are

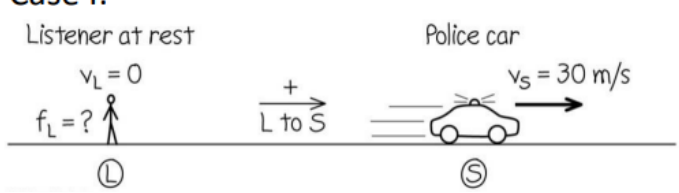
$$f_n = n\left(\frac{v}{2L}\right) = nf_1, \qquad n = 1, 2, \ldots$$



N=does not vibrate

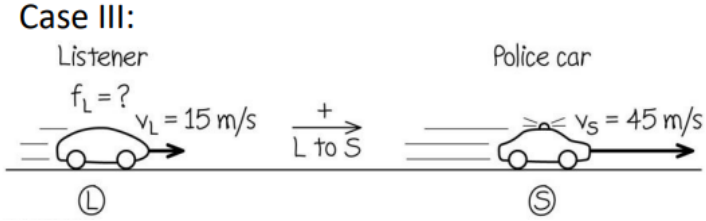
A police car's siren has frequency $f_S = 300$ Hz. Take speed of sound in still air, v, to be 340 m/s

Case I:



$$f_L = \frac{340\text{ m/s}}{340\text{ m/s} + 30\text{ m/s}}(300\text{ Hz}) = 276\text{ Hz}$$

Case III:



$$f_L = \frac{340\text{ m/s} + 15\text{ m/s}}{340\text{ m/s} + 45\text{ m/s}}(300\text{ Hz}) = 277\text{ Hz}$$